

Lab 2: Remote Patient Vitals Alert & Monitoring App

Jira Agile Backlog, Sprint Simulation & Analysis

Project Name	Remote Patient Vitals Alert & Monitoring App
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Problem Statement	#16 – Healthcare & Telemedicine

1. Project Overview

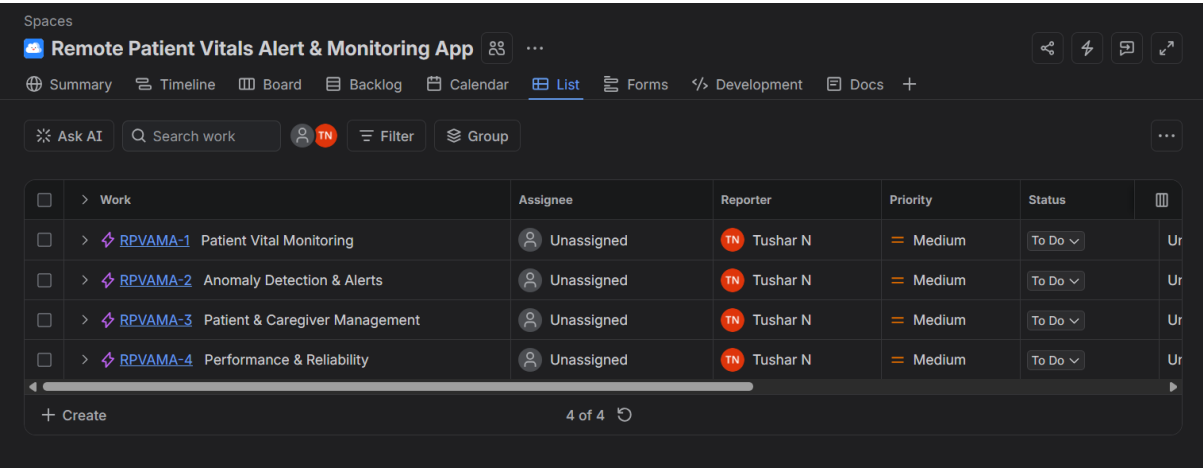
The Remote Patient Vitals Alert & Monitoring App is a continuous monitoring pipeline for post-operative patients. It ingests SpO2, heart rate and blood pressure telemetry, detects threshold breaches and escalates emergency alerts through a caregiver matrix. The target stakeholders are the Remote Patient and the On-Call Caregiver.

2. Epics and User Stories

Epic	User Story	Description	Priority	Story Points
Epic 1: Patient Vital Monitoring	US-01: Ingest Patient Vitals	As a system, I want to continuously receive SpO2, heart rate and BP data so that patient vitals can be monitored remotely.	High	8
Epic 1: Patient Vital Monitoring	US-02: View Current Vitals	As a caregiver, I want to view the patient's latest vital readings so that I can monitor their condition.	High	5
Epic 1: Patient Vital Monitoring	US-03: Configure Thresholds	As a caregiver, I want to configure clinical thresholds so that abnormal readings can be detected.	Medium	3
Epic 2: Anomaly Detection & Alerts	US-04: Detect Critical Anomalies	As a caregiver, I want the system to detect	High	8

		threshold breaches so that critical patient conditions are identified quickly.		
Epic 2: Anomaly Detection & Alerts	US-05: Generate Emergency Alert	As an on-call caregiver, I want to receive an immediate alert when a critical anomaly occurs so that I can take appropriate action.	High	8
Epic 2: Anomaly Detection & Alerts	US-06: Alert Escalation	As a caregiver, I want alerts to be escalated through the caregiver matrix when necessary so that emergencies are not missed.	High	5
Epic 3: Patient & Caregiver Management	US-07: Patient Monitoring Profile	As a caregiver, I want to access a patient's monitoring profile so that I can identify the patient associated with incoming vitals.	Medium	5
Epic 3: Patient & Caregiver Management	US-08: Caregiver Matrix	As an administrator, I want to maintain the caregiver escalation matrix so that alerts reach the appropriate caregiver.	Medium	5
Epic 4: Performance & Reliability	US-09: Real-Time Alert Processing	As a caregiver, I want critical anomalies to be processed within 2 seconds so that	High	8

		emergency situations can be addressed promptly.		
Epic 4: Performance & Reliability	US-10: Concurrent Telemetry Support	As a system administrator, I want the system to support at least 500 concurrent telemetry streams so that multiple patients can be monitored reliably.	Medium	13



3. Jira Epic View

Insert the Jira screenshot showing the four epics created for the project below.

[PASTE JIRA EPIC SCREENSHOT HERE]

4. Story Point Estimation & Planning Poker

Story points estimate relative effort, complexity and uncertainty. The Fibonacci-based values 3, 5, 8 and 13 were used for the user stories. Lower-point stories represent simpler monitoring or configuration activities, while alert processing, anomaly detection and concurrent telemetry support involve greater complexity and uncertainty.

5. Prioritized Backlog

Rank	User Story	Priority	Story Points
1	Ingest Patient Vitals	High	8

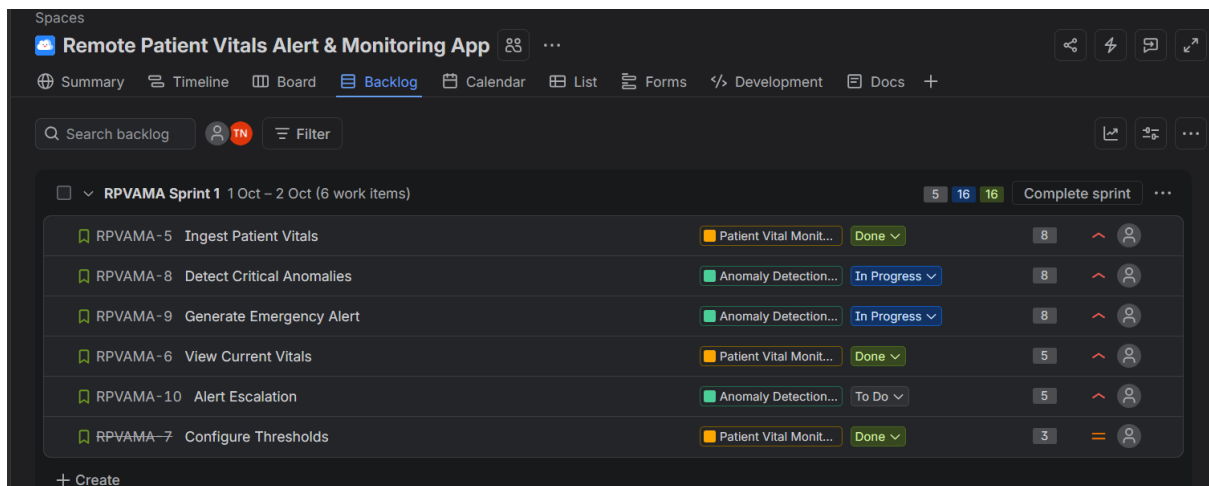
2	Detect Critical Anomalies	High	8
3	Generate Emergency Alert	High	8
4	Real-Time Alert Processing	High	8
5	View Current Vitals	High	5
6	Alert Escalation	High	5
7	Patient Monitoring Profile	Medium	5
8	Caregiver Matrix	Medium	5
9	Configure Thresholds	Medium	3
10	Concurrent Telemetry Support	Medium	13

6. Sprint Simulation & Burndown Charts

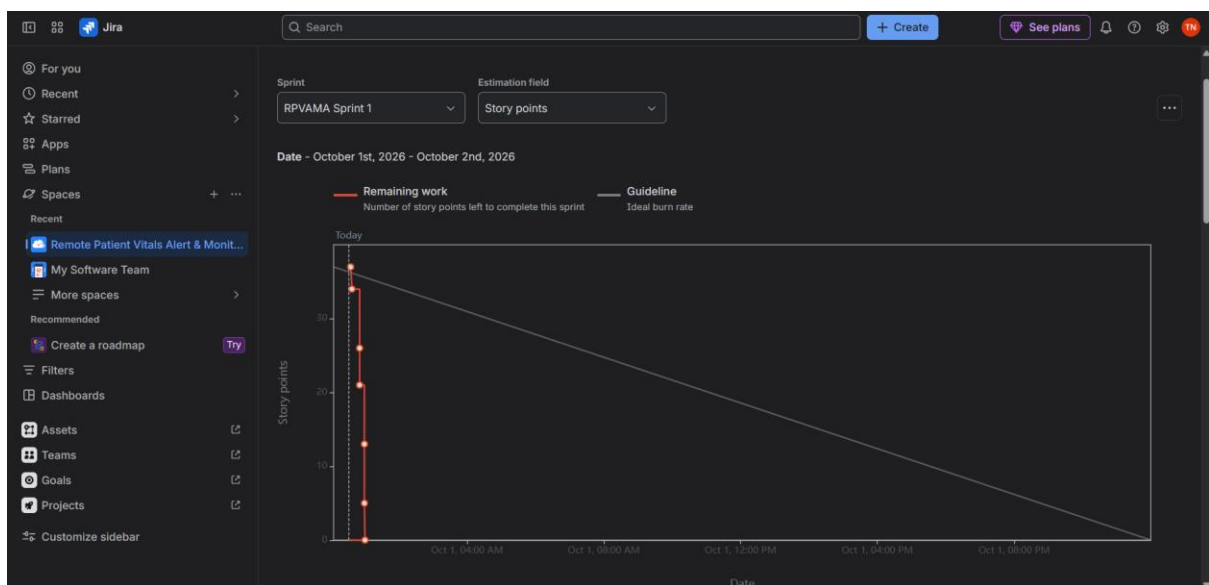
The backlog is organized into two mock sprints. Each sprint can be simulated through the To Do, In Progress and Done workflow states.

Sprint	User Stories	Total Story Points
Sprint 1	Ingest Patient Vitals; View Current Vitals; Configure Thresholds; Detect Critical Anomalies; Generate Emergency Alert	37
Sprint 2	Alert Escalation; Patient Monitoring Profile; Caregiver Matrix; Real-Time Alert Processing; Concurrent Telemetry Support	36

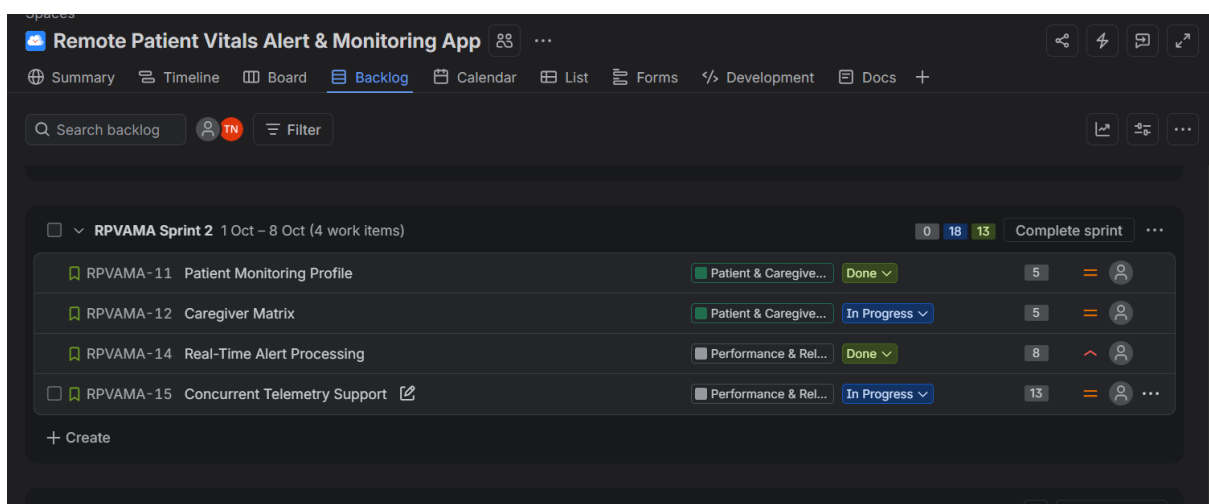
Sprint 1 Screenshot



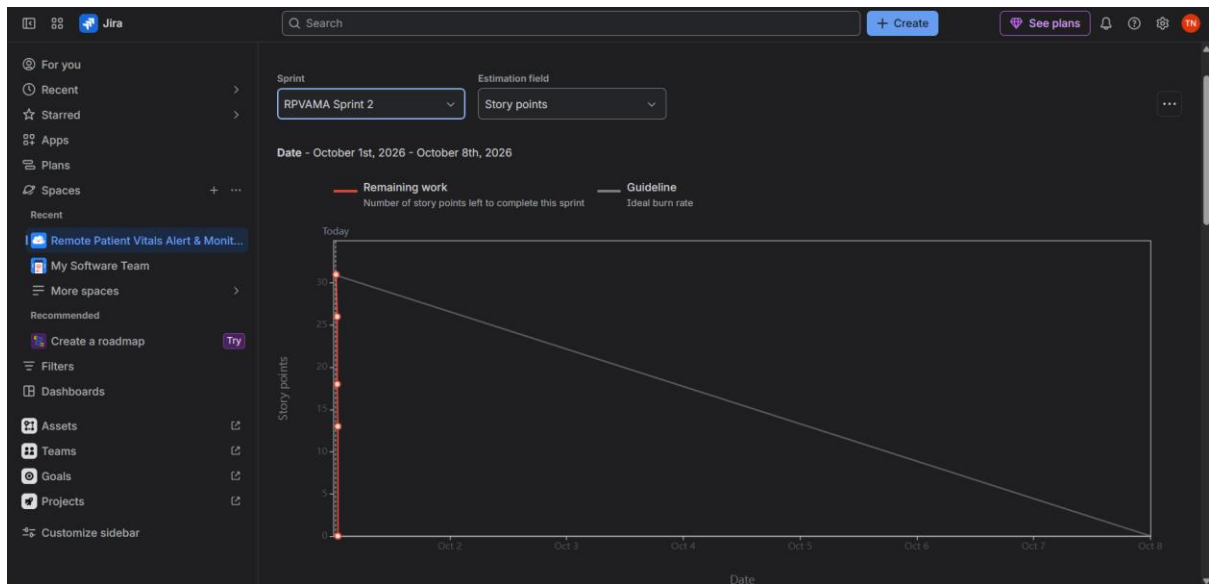
Sprint 1 Burndown Chart



Sprint 2 Screenshot

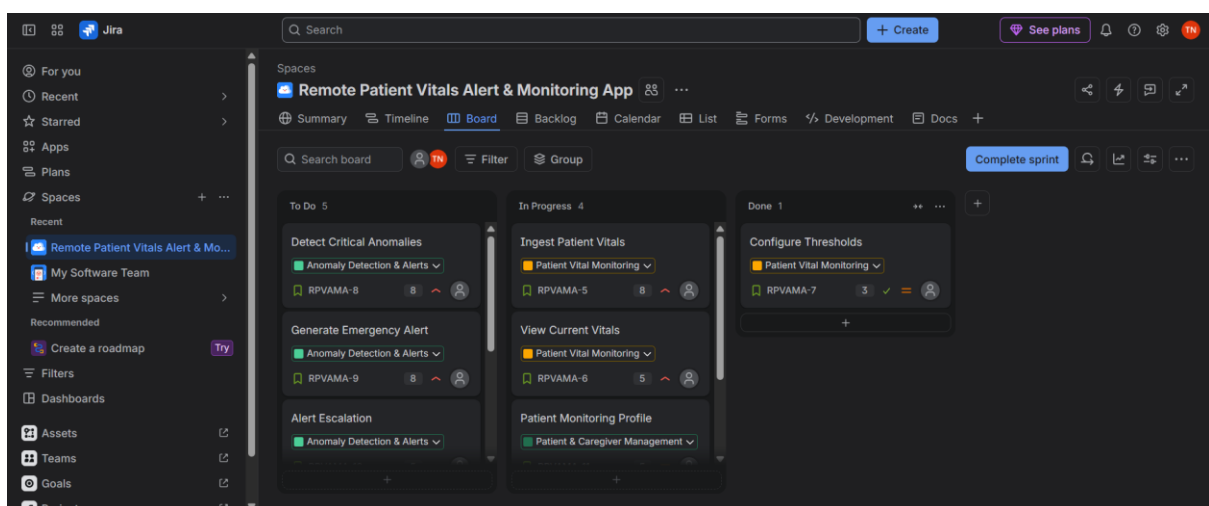


Sprint 2 Burndown Chart



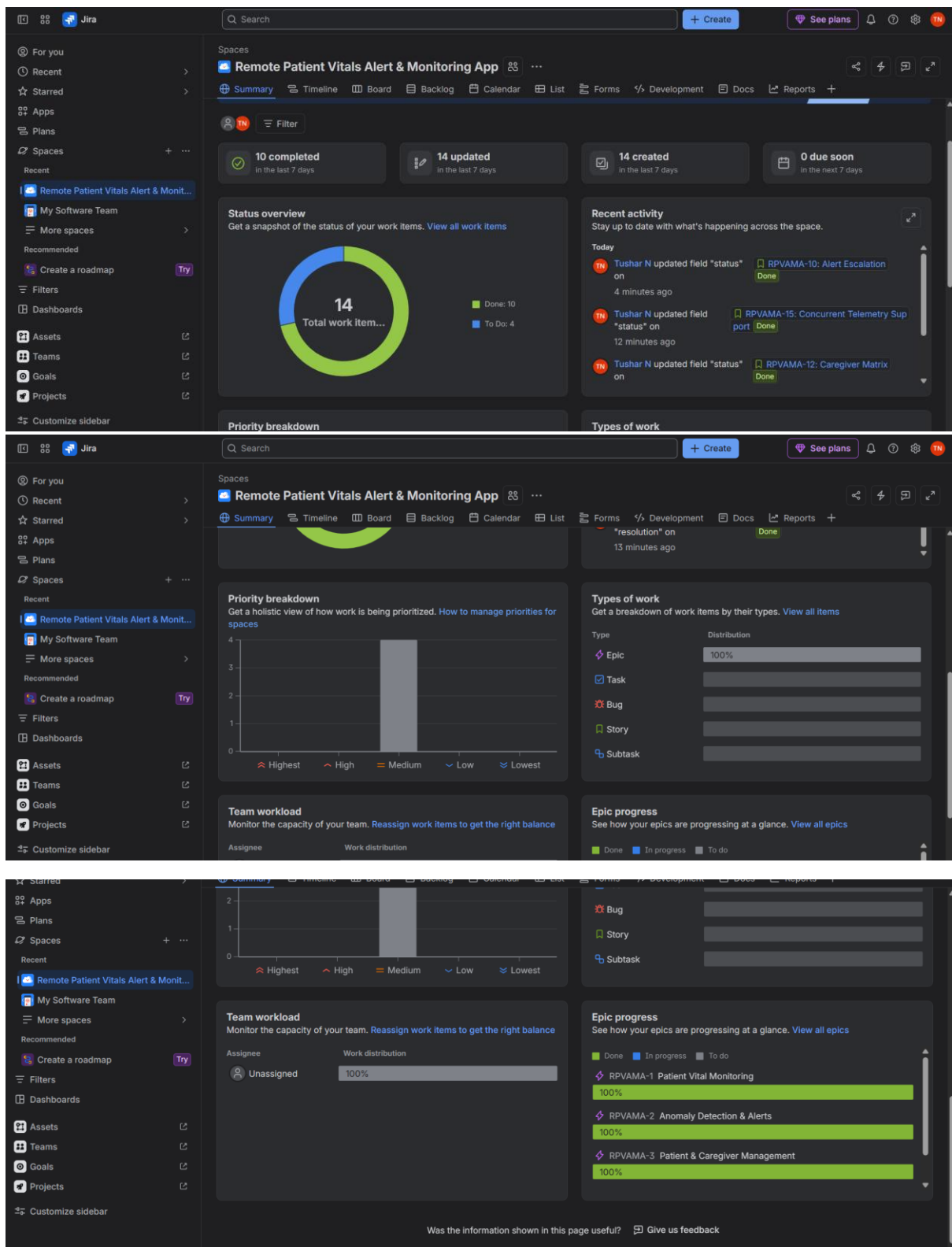
7. Active Sprint

The Active Sprint view shows the selected work items progressing through To Do, In Progress and Done. For the lab simulation, the stories are moved across the board to demonstrate Scrum execution.



8. Jira Summary Dashboard

The Jira Summary provides an overview of project work items, completion status, priority distribution, work types, team workload and epic progress. It provides visual evidence of the simulated Agile workflow.



9. Project Summary & Reflection

This Jira sprint simulation demonstrates the application of Agile Scrum practices to the Remote Patient Vitals Alert & Monitoring App. The requirements were divided into four Epics and ten User Stories covering patient vital ingestion, anomaly detection, emergency alerting, caregiver management, and performance requirements. Priorities and Fibonacci-based story points were assigned to organize and estimate the backlog.

The work was divided into two sprints so that the monitoring workflow could be developed incrementally. Sprint 1 focuses on receiving and viewing vitals, configuring thresholds, detecting anomalies and generating alerts. Sprint 2 focuses on escalation, caregiver management, real-time processing and concurrent telemetry support. The Active Sprint board and Burndown Charts provide visual evidence of work progression and remaining effort.

The Lab 1 requirements specify that critical anomalies should be flagged within 2 seconds and that the telemetry ingestion gateway should support at least 500 concurrent continuous telemetry streams with 99.99% uptime. These requirements are reflected in the corresponding Jira stories.